





Early–late asymmetry of the muon density with the UMD

Selection bias, core reconstruction, and how to recover it

Lautaro Silva Pizzi

with J. M. Figueira and F. Sánchez

ITeDA (CNEA-CONICET-UNSAM) — FCEN-UBA

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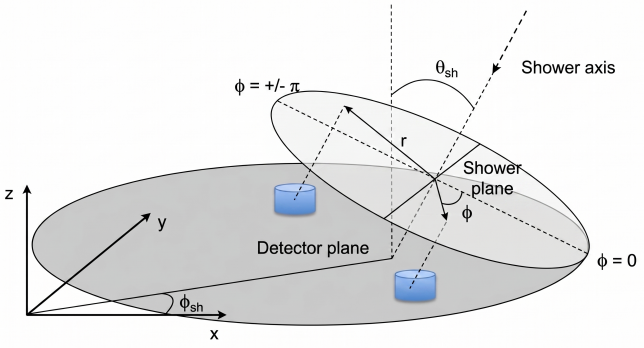
This talk is about the azimuthal asymmetry of the muon density, measured underground with the UMD. I'll show that the asymmetry is clearly there in simulations, then two analysis effects that can hide it or even flip its sign, and finally how we can account for one of them.

Transition: Let me first say what we measure.

The observable: first harmonic in the shower plane

- ϕ: station azimuth in the **shower plane**.
- Vertical shower: circular symmetry.
- Inclined shower: two distinct regions.
  - **early** ($\phi \approx 0°$): reached first, through *less* atmosphere.
  - **late** ($\phi \approx \pm 180°$): reached later, through *more* atmosphere.
- $A_1 > 0$: early excess; $A_1 < 0$: late excess.
- Fitted on an **ensemble** of showers, per (r, θ) bin.

$$\rho(r, \phi) = \langle \rho(r) \rangle \, (1 + A_1 \cos \phi)$$



The asymmetry of the total SD signal in inclined showers is well known. The question here is what happens with muons alone, which is what the UMD gives us: shielded, so free of the electromagnetic component. The observable is the first-harmonic amplitude A1 of the muon density in the shower plane. For a vertical shower the density is circularly symmetric. When the shower is inclined, two regions appear: the early side, reached first through less atmosphere, and the late side, reached later through more atmosphere. A positive A1 means more muons on the early side. A1 is fitted on an ensemble of showers, not event by event.

Transition: We start in the cleanest setup we have.

Why an early–late asymmetry?

Four mechanisms modulate the muon density in the shower plane:

Mechanism	Effect	Favours
Atmospheric attenuation	The late region is farther: more muons decay on the way	**early**
Detector response	Late muons arrive more inclined: a flat detector (UMD) sees a lower density there	**early**
Kinematic divergence	At the same r, the late region is farther ($1/d^2$ dilution) but needs a smaller opening angle (smaller, more probable p_t)	**early** / **late**
Geomagnetic field	Separates μ^+ and μ^-; second order for $\theta \lesssim 70^\circ$	—

A toy model combining them predicts $A_1 > 0$, growing with r and θ.

[20–30 s] Four mechanisms shape the muon density in the shower plane. Attenuation and the response of a flat detector favour the early side, the kinematic divergence pulls both ways, and the geomagnetic field is second order at these angles. A toy model combining them predicts a positive A1, growing with distance and zenith angle.

Transition: Now the simulations.

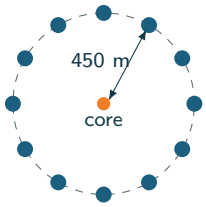
The reference: Monte Carlo geometry

Simulations and the Dense Ring

- Official CORSIKA + *Offline* production, with SD-750 and UMD:

Hadronic models	SIBYLL 2.3e (main), QGSJET-III.01
Primaries	p (main), He, Fe
Energy	$10^{17.5}$–10^{18} eV
Zenith angle	$\theta \leq 65^\circ$

- **Dense Ring**: 12 virtual stations at $r = 450$ m from the **true** axis, every 30° in the shower plane. Same r for all: the radial fall-off of the density cannot produce a modulation in ϕ.

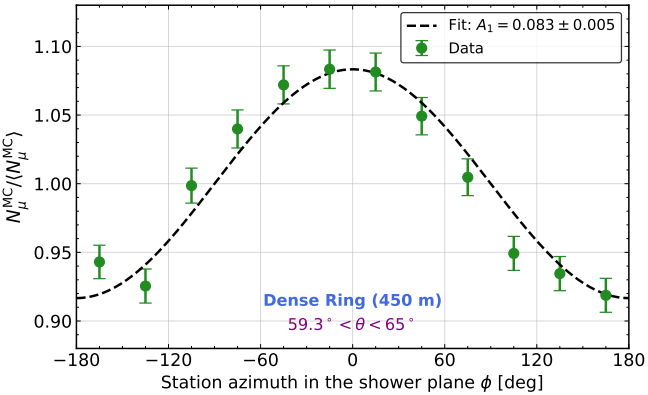


Shower plane.

[20–30 s] Official collaboration production; most of what I show is protons with SIBYLL 2.3e. We start with the Dense Ring: twelve virtual stations at 450 meters from the true axis, so the fall of the density with distance cannot fake a modulation.

Transition: This is what the muon density looks like on the ring.

A clear modulation on the Dense Ring

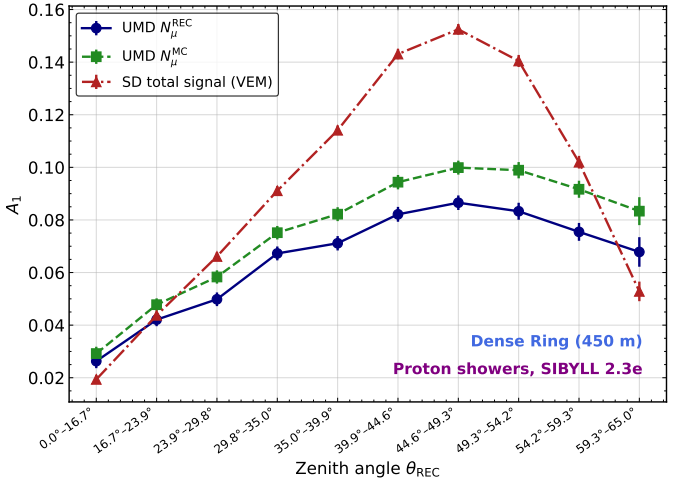


- Injected muon number N_μ^{MC}, $59.3^\circ < \theta < 65^\circ$.
- Maximum in the **early** region ($\phi \approx 0$).
- Early–late difference $\approx 17\%$ peak to peak ($\pm 8\%$ around the mean).

Here are the twelve ring stations for the most inclined bin, using the injected muon numbers. The density peaks on the early side and the first harmonic gives an amplitude of 0.083: the early side has about eight percent more muons than the mean, the late side eight percent fewer. With the reconstructed muon numbers the same bin gives 0.068; that difference is the counting reconstruction, in the backup. (If asked: the Dense Ring zenith bins are in reconstructed zenith angle.)

Transition: How does it change with zenith angle?

Comparison: SD vs. UMD on the Dense Ring



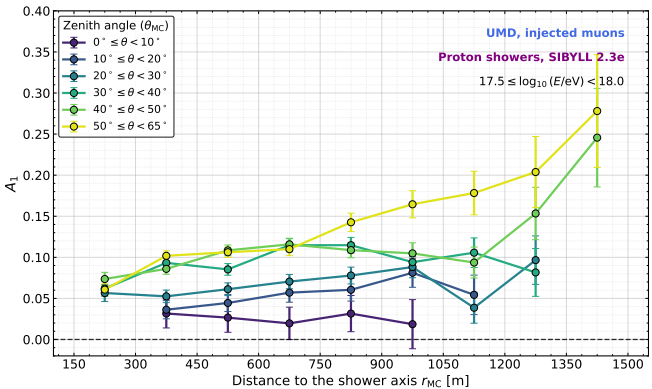
- UMD: $A_1 > 0$, grows with θ up to $\sim$ 45–50°.
- UMD: lower amplitude with reconstructed (N_μ^{REC}) than with injected (N_μ^{MC}) muons, same shape.
- SD total signal: larger than the UMD at $\sim$ 20–60° (EM-dominated); drops sharply above $\sim$ 50°.
- SD below the UMD only at $\theta \gtrsim 60°$.

[20–30 s] The muon asymmetry grows with zenith angle. The SD total signal is larger, because it is dominated by the electromagnetic part, and falls below the UMD only above about sixty degrees.

Transition: For the radial dependence we need the real Infill array.

UMD radial dependence on the Infill (MC geometry)

Real Infill stations (750 m): the distance to the axis varies from station to station $\Rightarrow A_1(r)$.
Station r and ϕ depend on where the axis is: here the **true** core and direction.



- **UMD only** (injected muons).
- Radial bins of $\Delta r = 150$ m; zenith bins of 10°.
- $\theta < 10^\circ$: compatible with zero.
- 10–40°: grows with r, then **flattens**.
- $\theta \geq 40^\circ$: **grows with** r.

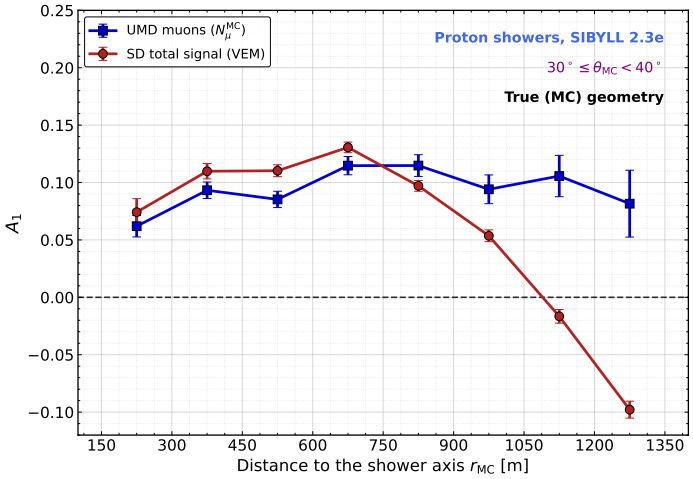
This is our reference.

Still simulations, but now with the real Infill stations. Their distances to the axis vary, so we can follow A1 as a function of r. The distance and azimuth of each station depend on where the shower axis is: we can use the true core and direction from the simulation, or the reconstructed ones, and the main difference between the two is the core position. Here we use the true one. This is the UMD alone, with the injected muon numbers, in radial bins of 150 meters and zenith bins of ten degrees. Near-vertical showers show no significant asymmetry. Between ten and forty degrees it grows with distance and then flattens. Above forty degrees it grows with distance. (If asked: at forty to fifty degrees it stays around 0.1 and rises only beyond about twelve hundred meters, where the statistics are lower.) This is what an ideal analysis should find.

Transition: Now we compare with the surface detector in the same showers.

The SD inversion and the selection bias

The SD comparison: inversion far from the core

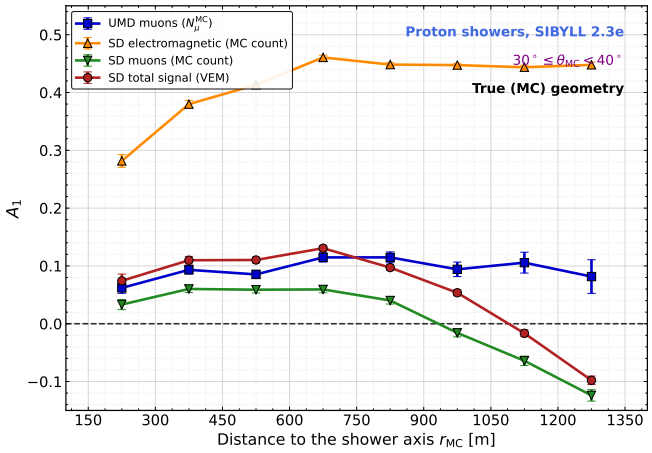


- We now compare the UMD and SD asymmetries in the same Infill showers.
- MC geometry, $30^\circ \leq \theta < 40^\circ$.
- UMD: positive at all r.
- SD total: a maximum, then negative far from the core.

Now we compare the UMD with the surface detector, on the same Infill showers and with the same true geometry. The UMD stays positive at all distances. The total SD signal peaks, then decreases, and far from the core becomes clearly negative, an excess on the late side. Since the core and direction are the true ones, this cannot be a reconstruction error.

Transition: Which part of the tank signal is responsible?

It is the SD muon component that inverts



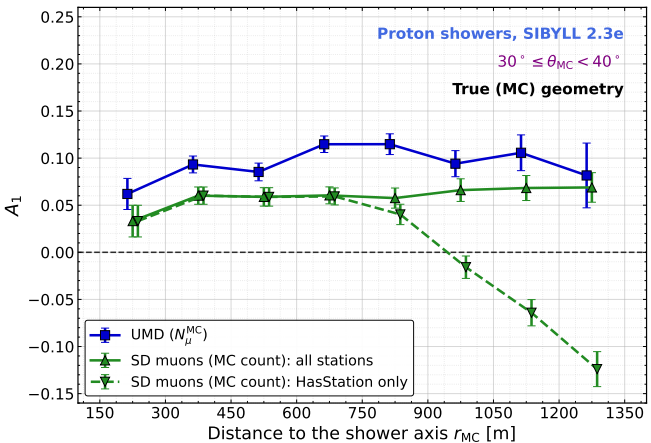
- MC particle counts in the tank, split by component.
- **EM**: positive at all r.
- **SD muons**: positive near the core, negative beyond $\sim$ 900 m.
- The SD total follows its muons at large r.

Same quantity, opposite sign in UMD and SD.

Splitting the tank signal with the simulation truth, the electromagnetic part is positive everywhere. The muon part is what inverts, and the total SD signal follows it far from the core. So we compare the same physical quantity, muons crossing a detector, and get opposite signs in the UMD and in the SD. Our analytic model of muon propagation does not produce a negative asymmetry at these distances, so we looked at how the observable is built.

Transition: The key point: we never average over all stations.

Without `HasStation`, the inversion disappears



- Same MC counts, coordinates and fit: only **which stations** enter changes.
- All stations: SD muons **positive at all** r ($+0.03$ to $+0.07$).
- Same sign and trend as the UMD, **but not the same values**.

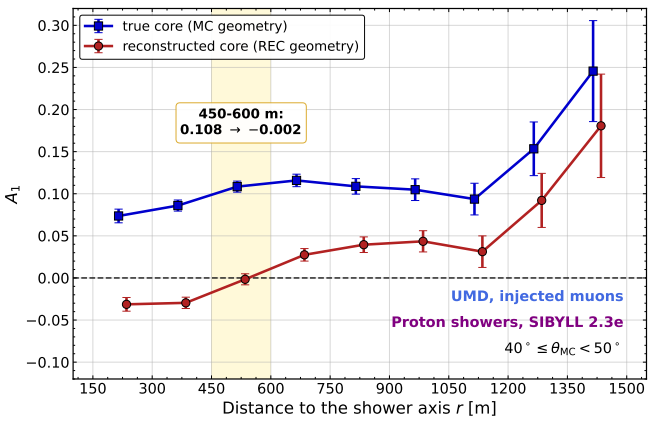
Every curve so far is an average over the stations that belong to the reconstructed event. If the chance of being kept depends on the station's own muon content, and differently on the early and late sides, the averages are biased. The direct test: same Monte Carlo muon counts in the tanks, same coordinates, same fit; only which stations enter changes. Over all simulated stations the SD muon asymmetry is positive at all distances, between plus three and plus seven percent. Requiring the station to be in the reconstructed event changes nothing up to about 750 meters, where practically every station is kept, and then drives the asymmetry down to minus 0.12 at twelve to thirteen hundred meters: plus 0.069 plus or minus 0.016 over all stations, against minus 0.124 plus or minus 0.019 with the requirement. The errors come from a bootstrap over the 960 showers, keeping the stations of each shower together. So the sign flip is a property of the station sample. Without the selection the SD muons have the same sign and trend as the UMD, but between 150 and 900 meters the UMD is still 0.03 to 0.06 higher, and that difference excludes zero at 95 percent (backup slide); detector response and threshold are candidates, this is open.

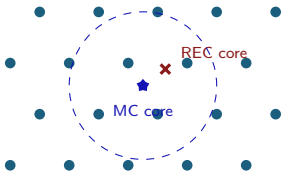
(If asked whether it is the trigger: the excluded tanks hold at most one muon in 93 to 95 percent of the cases, backup.)

Transition: So far, true core and direction. In real data we only have the reconstructed ones.

Reconstructed geometry and the core position

REC geometry suppresses the asymmetry near the core



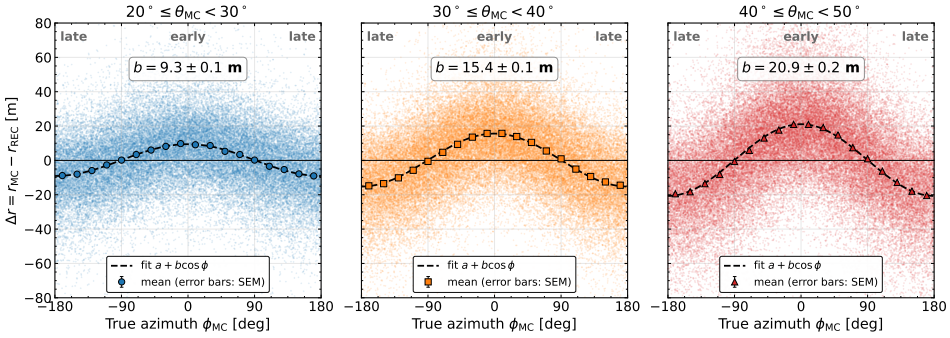


- Same UMD modules and **injected** muons; **only the core changes**.
- Near the core A_1 drops to **zero or below**; farther out the curves **converge**.
- The effect comes from the **offset between the true and the reconstructed core positions**.

This is the reference curve of slide eight, forty to fifty degrees, with the same UMD modules and the same injected muon numbers, now computed a second time with the reconstructed core. Near the core the asymmetry drops to zero or becomes negative: at 450 to 600 meters it goes from 0.108 to minus 0.002. Farther out the two curves approach each other: the loss is not global, it shrinks with distance. We also swapped the two quantities one at a time: using the reconstructed or the true azimuth gives the same A1 within 0.002, so everything comes from the distance to the reconstructed core. (If asked, other zenith bands at 450 to 600 meters: 0.061 to 0.011 at twenty to thirty degrees, 0.085 to 0.002 at thirty to forty.)

Transition: So where is the reconstructed core?

The reconstructed core is shifted toward the early region



- $\Delta r = r_{\mathrm{MC}} - r_{\mathrm{REC}}$ vs. ϕ is fitted by $a + b\cos\phi$, the modulation expected from a **rigid displacement of the core** by b: early stations are placed **closer** to the core, late ones **farther**; $a \simeq 0.2$ m.
- Likely cause: the reconstruction LDF is **symmetric in** ϕ; it absorbs the early excess by **moving the core toward the early side, where there is more signal**.

For each station we compare its distance to the true and to the reconstructed core, as a function of its true azimuth. If the core error were random, this would be flat. Instead it follows a cosine: early stations are placed closer to the core than they are, late stations farther. That is the signature of a systematic shift of the reconstructed core toward the early side: if the core is displaced rigidly by b toward the early direction, to first order every station's distance changes by b cos phi, which is exactly the fitted form. The constant a comes out at about 0.2 meters, so there is no azimuth-independent radial bias. The error bars are the standard error of the mean in each azimuth bin, smaller than the markers; the cloud behind them is the station-by-station spread, about twenty meters. So the modulation is very well determined even though single stations scatter a lot. The fitted shift is 9.3, 15.4 and 20.9 meters, with errors of 0.1 to 0.2 meters. (If asked: those errors treat the stations as independent, but the stations of one shower share the same core error, so they are underestimated; a bootstrap over events is pending. The trend with zenith angle is far larger than any plausible error.) A likely reason: the LDF used in the reconstruction is symmetric in azimuth. Faced with more signal on the early side, the only thing the fit can do is move the core toward it. (Infill stations with reconstructed distance below 1400 meters.)

Transition: What does a shift of this size do to the LDF, and to A1?

A shifted core adds a $\cos\phi$ term to the LDF

- Binned by r_{REC}, a station at r is really at $r + b\cos\phi$: its density belongs to **another radius**.

$$\rho(r + b\cos\phi) \simeq \rho(r)\Big[1 - \beta\,\frac{b}{r}\cos\phi\Big] \;\Rightarrow\; \boxed{\Delta A_1 \simeq -\beta\,\frac{b}{r}} \qquad \beta = -\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r}\ \text{(local)}$$

$450 \le r < 600$ m, injected muons

θ	b [m]	β	ΔA_1 measured	$-\beta b/r$ predicted	residual
20–30°	9.3	2.96	−0.050	−0.052	+0.002
30–40°	15.4	2.72	−0.084	−0.080	−0.004
40–50°	20.9	2.52	−0.110	−0.100	−0.010

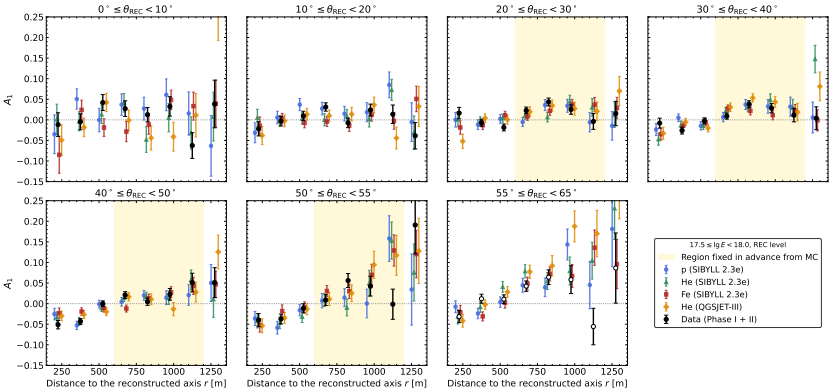
[20–30 s] A shifted core puts each station at the wrong distance, and since the LDF is steep this adds a cos phi term of opposite sign: Delta A1 is about minus beta b over r. With b and beta measured on their own, the prediction matches the loss within 0.01.

Transition: Does this pattern show up in field data?

A first look at field data

Field data follow the REC-geometry simulations

FIELD DATA – PRELIMINARY



- Data (Phase I + II) and simulations with the **same reconstruction**, $17.5 \leq \lg E < 18.0$.
- Same radial pattern: $A_1 \leq 0$ near the core in inclined showers, positive farther out: the **signature of the core shift**, seen in data.

[20–30 s] Field data, in black, follow the simulations processed with the same reconstruction, including the negative asymmetry near the core in inclined showers: the core-shift signature, seen in data. (If asked for a single number: in the best Phase II cell A1 $= 0.063$ plus or minus 0.022, 2.8 sigma, backup.)

Transition: Let me summarize.

Conclusions

Conclusions

1. The UMD sees a **positive early–late muon asymmetry**.
2. The SD inversion far from the core is **compatible with a station-selection effect**.
3. The reconstructed core is **shifted toward the early region**, which suppresses A_1 near the core.
4. The loss **is predicted by** $\Delta A_1 \simeq -\beta\, b/r$.
5. Field data **follow the REC-geometry simulations** (preliminary).

Five points. The muon asymmetry is there. What we measure depends on how we build the observable: the SD inversion is compatible with a selection of the stations we average over, and the reconstructed core can erase the asymmetry near the core. The second effect is predicted by beta b over r, and the field data already show its signature. Next: parametrize b and beta with zenith angle and validate them on independent simulations, correct the field data, and separate trigger, readout and reconstruction inside HasStation.



Thank you!

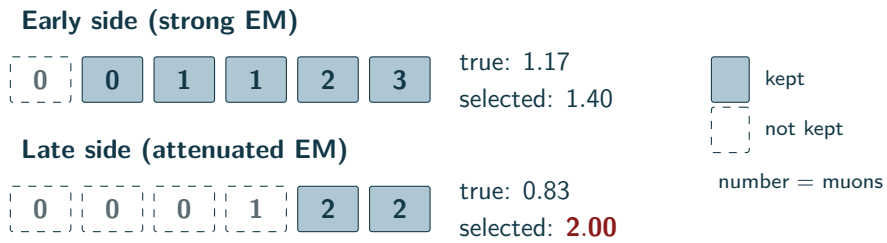
Questions?

Image: A. Chantelauze, S. Staffi, L. Bret (Pierre Auger Observatory)

Backup: the station selection biases the average

HYPOTHESIS

A station enters the analysis only if it **belongs to the reconstructed event** (`HasStation`). If the probability that a station is kept depends on its own muon content, and this dependence is not the same on the early and late sides, the **regional averages are biased**.



Both averages go up, the late one much more: the sign flips **without moving a single muon**.

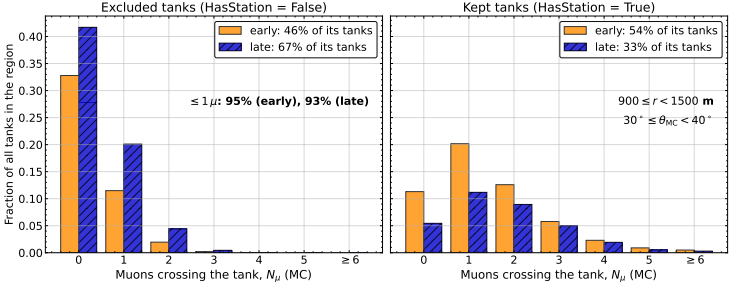
Illustrative cartoon. In the simulation, at 900–1500 m, 54% of the early and 33% of the late tanks are kept.

Every curve so far is an average over stations that are part of the reconstructed event. For the UMD that seemed natural, since it has no trigger of its own. But suppose the chance that a station is kept depends on how many muons it has. On the early side the electromagnetic signal is strong, so almost every station is kept, even one with no muons; only a few muon-poor ones are lost and the average goes up a little. On the late side the electromagnetic signal is weaker, so mostly the stations with more muons survive and the average goes up a lot. In this example the true averages are 1.17 and 0.83, an early excess; the selected ones are 1.40 and 2.00, a late excess. The sign flips although no muon was created or moved. This is a hypothesis; the next two slides test it with the simulation.

Transition: The direct test is to remove the requirement.

Backup: the excluded tanks hold very few muons

- If the inversion comes from the **trigger**, the tanks dropped by the reconstruction (`HasStation` = False) should hold, on average, at most one muon: too few for the station to trigger.



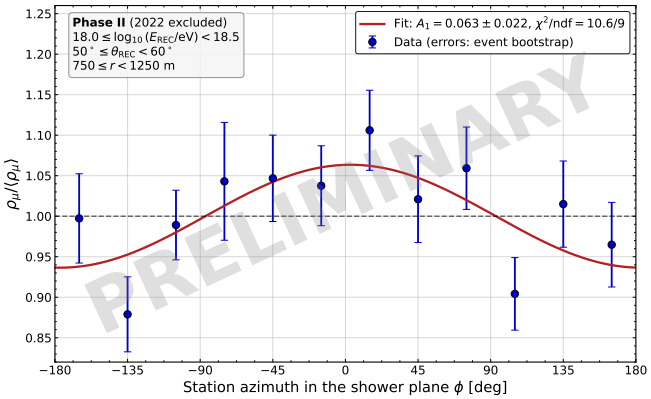
	$\langle N_\mu \rangle$ all tanks	$\langle N_\mu \rangle$ kept tanks	fraction kept
early	0.96	1.49	0.54
late	0.87	1.69	0.33

If the inversion is a trigger effect, there is a clear prediction: the tanks that the reconstruction drops should have been crossed by at most about one muon. If we found excluded tanks with several muons, a trigger explanation would not be credible. So we count the Monte Carlo muons in the tanks between 900 and 1500 meters. Each region is normalised to all of its tanks, so the area of the bars is the fraction lost or kept: on the early side 46 percent of the tanks are dropped, on the late side 67 percent. Ninety-five percent of the excluded early tanks and ninety-three percent of the late ones have at most one muon. The table is the bias in numbers: over all tanks the early side has more muons, 0.96 against 0.87, which is the true positive asymmetry; over the kept tanks it is the other way round, 1.49 against 1.69. (If asked why the kept late tanks have more muons than the kept early ones: the selection raises both means, but much more on the late side, by a factor 1.9 against 1.55, because there the electromagnetic signal is weaker, about half, 14 against 28 particles in the excluded tanks, so only muon-rich tanks pass; on the early side tanks with zero or one muon are often kept thanks to their electromagnetic signal. At 1200 to 1350 meters only 18 percent of the late stations are kept, against 42 percent of the early ones.) This supports the trigger picture, but HasStation also includes readout and reconstruction, and we have not separated them, so it is consistent with a trigger effect, not a proof. (Early: $|\phi| < 60°$, late: $|\phi| > 120°$; MC geometry.)

Transition: So far, true core and direction. In real data we only have the reconstructed ones.

Backup: the modulation in one field-data cell

FIELD DATA – PRELIMINARY

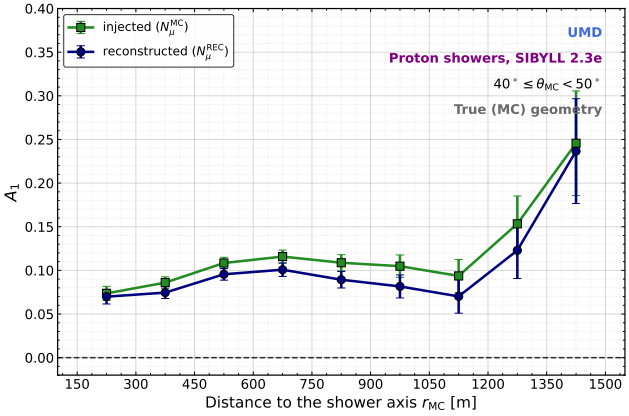


- UMD muon density vs. ϕ, normalised to its mean, Phase II; REC geometry.
- Maximum at $\phi \approx 0$: **early excess**, as in the simulations.
- $A_1 = 0.063 \pm 0.022$ (2.8σ), $\chi^2/\mathrm{ndf} = 10.6/9$; errors from a bootstrap over **events**.
- **Preliminary**: no core-shift correction, no systematics, no MC above 10^{18} eV.

This is the UMD muon density measured in the field, as a function of azimuth in the shower plane, for Phase II events in the cell shown in the box. The maximum is on the early side, as in the simulations. The first harmonic gives 0.063 plus or minus 0.022. That error comes from a bootstrap over events, because the modules of one shower are correlated; treating every module as independent gives 0.017, which overstates the significance. So this is a 2.8 sigma early excess, with a good fit. It is preliminary: the core correction we just discussed is not applied, there is no systematics study, and our simulations stop at ten to the eighteen. (If asked: part of the zenith range, 55 to 60 degrees, is outside the standard SD-750 energy calibration; and 2022 Phase II commissioning data are excluded.)

Transition: And the radial pattern?

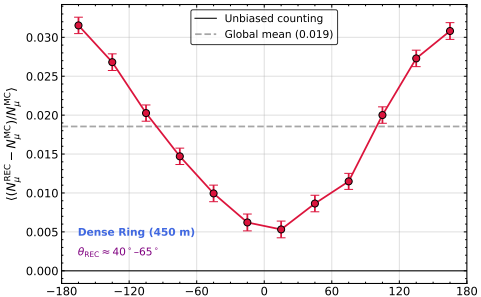
Backup: muon-counting reconstruction (MC geometry)

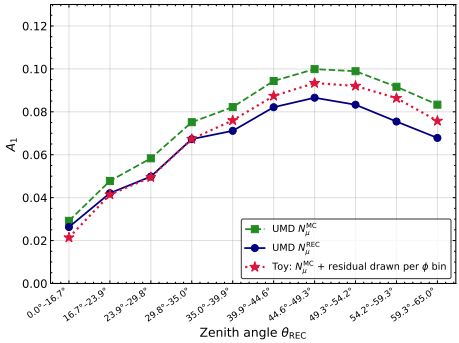


- 40°–50°, true core and direction.
- The reconstructed muon number lowers A_1 by 5–23%, with the same shape.
- A separate effect from the core shift.

Backup: the counting bias depends on azimuth

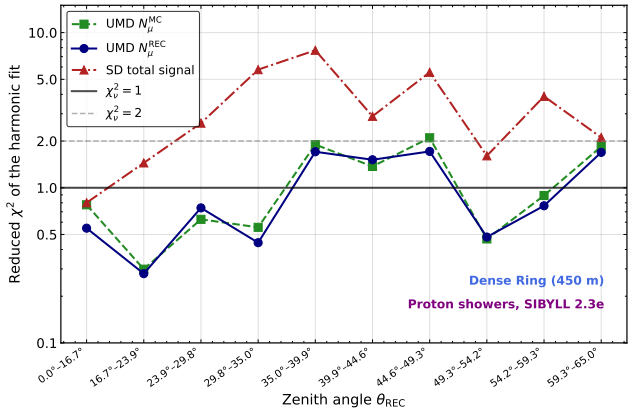
SIMULATIONS – DENSE RING





- Left: mean relative residual $(N^{REC} - N^{MC})/N^{MC}$ vs. ϕ: ≈ 0.005 early, ≈ 0.03 late. More over-counting where there are fewer muons $\Rightarrow$ lower A_1.
- Right: injecting the residuals drawn per ϕ bin into N_μ^{MC} reproduces A_1^{REC} up to $\theta \approx 35^\circ$; above, half to two thirds of the loss. The remnant is open.

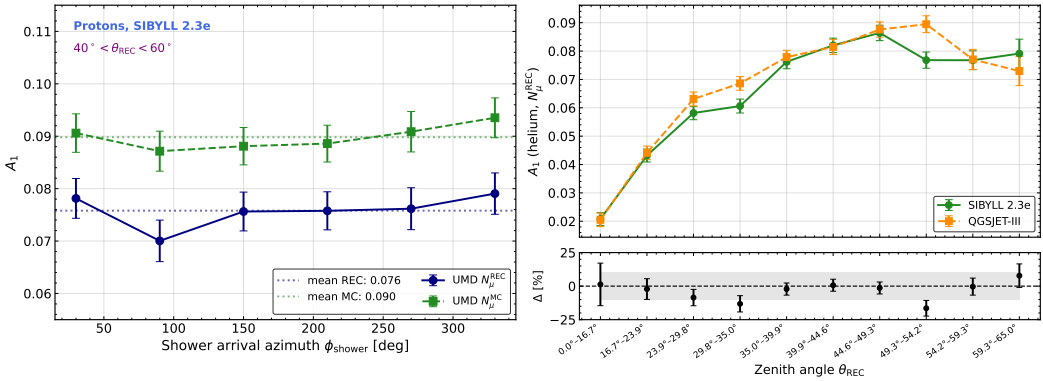
Backup: fit quality on the Dense Ring



- UMD, MC and REC: $\chi^2_\nu \approx 1$ (0.3 to 2.1).
- SD: χ^2_ν up to ~ 8. With such small errors the fit is sensitive to higher harmonics and to correlations between stations of one shower.

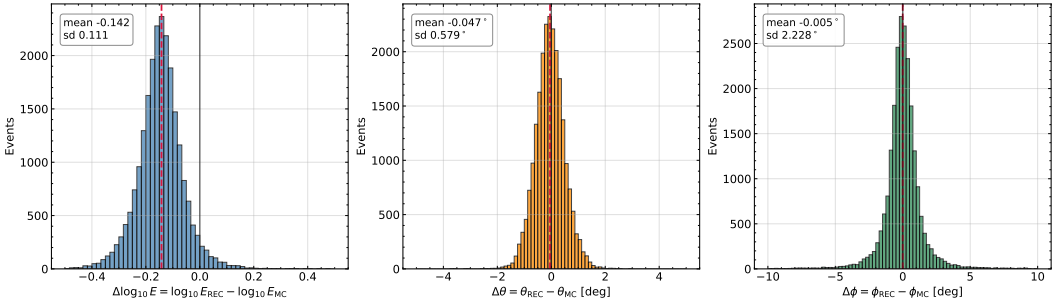
Backup: robustness

SIMULATIONS – DENSE RING



- Left: A_1 does not depend on the arrival azimuth of the shower.
- Right: helium, SIBYLL 2.3e vs. QGSJET-III: within $\pm 10\%$ over most of the range.

Backup: reconstruction performance



- Energy: mean -0.142, spread 0.111 in $\Delta \log_{10} E$ (fewer muons in the simulation than in data, with a data-based energy calibration).
- θ: bias $-0.05°$, spread $0.58°$. ϕ: bias $-0.005°$, spread $2.2°$.

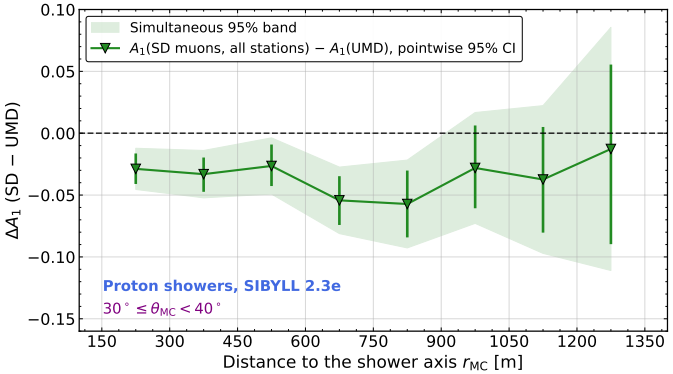
Backup: the core shift and the Dense Ring

- In reconstructed coordinates the ring **is** eccentric, but the stations are physically at 450 m from the **true** axis, and they are not re-binned in r.
- The assigned azimuth rotates by $\sim b/r \approx 2^\circ$, much less than the 30° spacing; the assigned radius changes by at most $b \approx 20$ m.
- Both affect N_μ^{MC} and N_μ^{REC} in the same way, so they cannot explain the MC–REC gap on the ring (that gap is the counting bias).

Backup: is the UMD biased by the selection too?

- The UMD has no trigger of its own: a module is read only if the tank of its station is kept. So the UMD curve is also averaged over `HasStation` stations.
- But the selection acts on the **tank's own particles** (its muons and its EM signal). The module counts **other muons**: a different area, not right under the tank, behind $\sim$2.3 m of soil ($\sim$1 GeV threshold).
- Given the station position, the expected UMD count is still the local muon density: to first order, **no bias**.
- A residual effect can enter through what tank and module share, the fluctuation of the local density from shower to shower and station to station. Not quantified here; the UMD count of the stations that were not kept is not stored in this production.

Backup: SD muons over all stations vs. UMD



- Without `HasStation`, the SD muons stay **below** the UMD: $\Delta A_1 \approx -0.03$ to -0.06 between 150 and 900 m.
- There the pointwise 95% interval excludes zero; beyond 900 m the difference is not resolved.
- Paired bootstrap over the 960 showers (UMD and SD from the same showers).
- Candidates: flat plate vs. tank response, UMD threshold. Open.